Kalasalingam Academy of Research and Education

OPENSCOUT – AI POWERED OPEN SOURCE PROJECT FINDER

Project ID: PRJ051

Team ID: T051

Team GitHub Repository: https://github.com/Thejes-reddy-1264/OpenScout---AI-powered-open-source-project-finder

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Project Trainer / Guide

Mr. SHIRWIN PRINCE I

Table of Contents

1. Introduction ..... Section 1
2. Problem Statement & Objectives ..... Section 2
3. System Architecture (RAG Pipeline) ..... Section 3
4. Implementation Details & Tech Stack ..... Section 4
5. Results and Testing ..... Section 5
6. Conclusion & Future Scope ..... Section 6

1 Introduction

Open source software has become an important part of the modern software development industry. With platforms like GitHub there are thousands of repositories available for you to look at the source code to increase your knowledge and understanding of new technologies, improve your programming skills and collaborate on open source projects. However, finding the correct open source project can be difficult for beginner and intermediates. Current methods for searching for repositories rely heavily on keywords, programming languages and popularity. This means that the user's technical skills, interests, experience and what type of contribution they want to make may not be taken into consideration.

The Open Source Project Finder is an intelligent system that helps you find the right open source repositories. It uses the information about the user's preferences and repository to suggest the right open source projects for them. The system will utilise data from GitHub repositories, intelligent search, filtering and recommendations to find the correct projects. queries.

2 Problem Statement & Objectives

Problem: Developers, especially beginners, often struggle to identify suitable open-source projects for contributing and improving their technical skills. Searching through numerous GitHub repositories manually is time-consuming. Furthermore, developers may find repositories that are either too complex for their experience level or unrelated to their interests.

Traditional keyword-based searches do not always provide personalized recommendations based on skills, preferred technologies, experience, and contribution goals.

Objectives:

- To develop a centralized platform for discovering open-source GitHub repositories.
- To recommend projects based on users' programming skills and technology preferences.

- To integrate GitHub repository information for displaying relevant project details.
- To provide advanced filtering based on programming languages, topics, stars, and experience levels.
- To help beginners identify repositories suitable for their technical knowledge.
- To simplify the process of finding beginner-friendly issues and contribution opportunities.
- To provide an intuitive interface for searching, exploring, and saving interesting repositories.

3 System Architecture

The system consists of 3 main phases: Data Collection & Processing, Project Recommendation and Query Processing & Results. Firstly collects data from repositories on GitHub, then processes the data to understand users preferences and recommend them suitable open source projects.

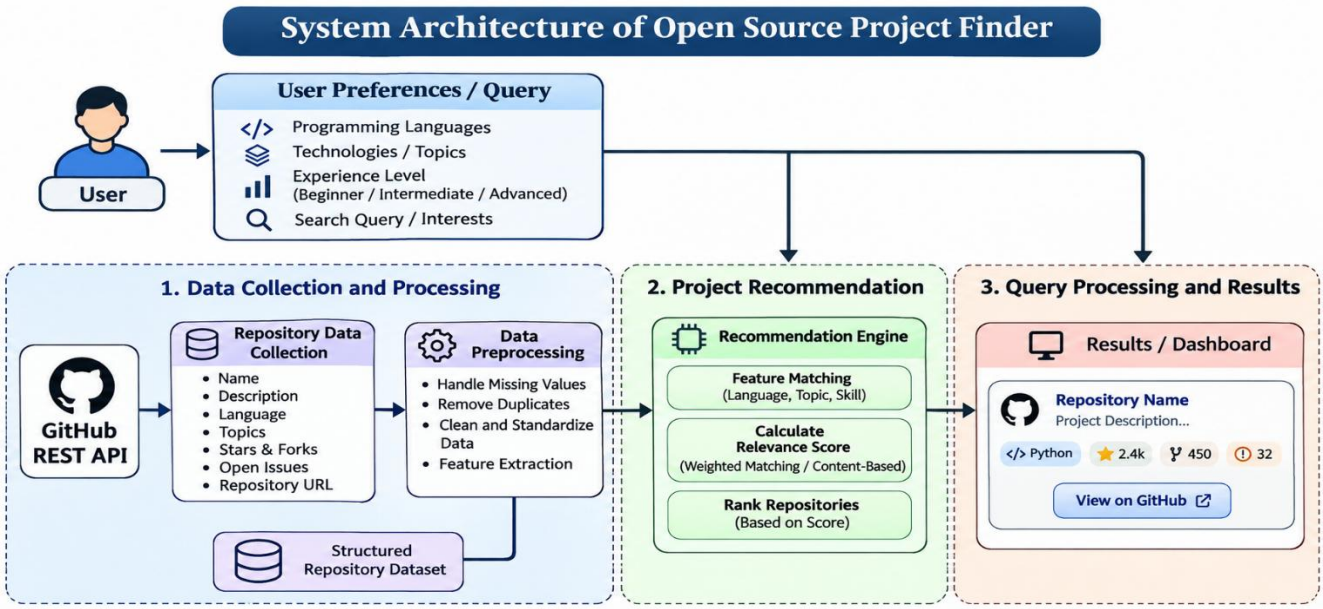


Figure 1: System Architecture of Open Source Project Finder

3.1 Data Collection and Processing Phase

The System gather system gathers data from GitHub repositories using the GitHub REST API. This includes: Repository name, Repository description, Repository programming language, Repository topics, Number of stars, Forks, Open issues, and Repository URL.

This data is then preprocessed and structured. Data preprocessing includes dealing with missing data, removing duplicates, and using feature extraction to provide features that can be used to find characteristics of repositories, which will help with providing quality recommendations for projects. The information is stored so it can be efficiently searched through.

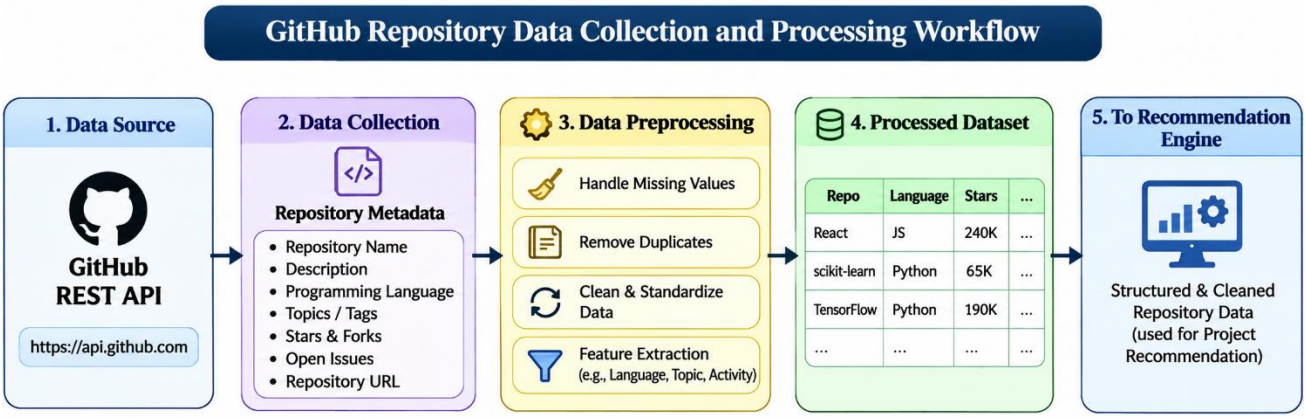


Figure 2: GitHub Repository Data Collection and Processing Workflow

3.2 Project Recommendation Phase

The recommend module takes into account the technical preferences of the user such as programming languages, frameworks, interests, skills and experience level. This is done via a recommend mechanism.

A weighted feature match or content based filter is used to compute a score for each repository in terms of the compatibility of languages, technology related topics, skills needed and how active the repository is. This allows the repositories to be ranked based on relevance and help the user find projects that suit their technical knowledge and interests.

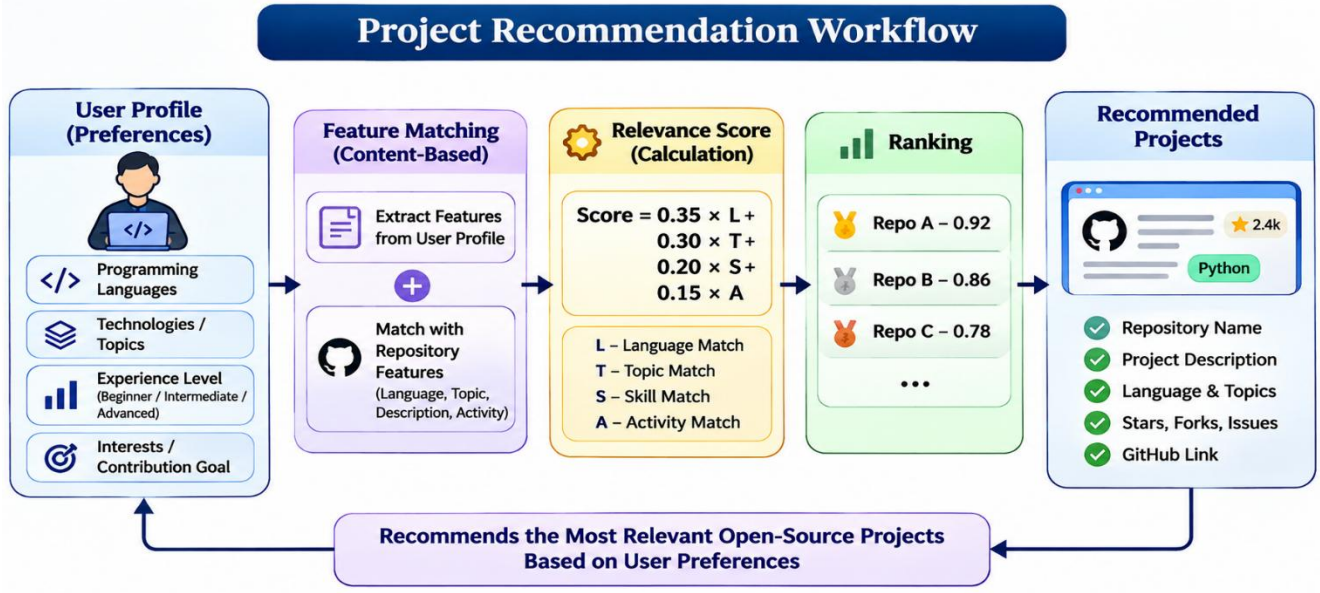


Figure 3: Personalized Open-Source Project Recommendation Workflow

3.3 Query Processing and Results Phase

Upon receiving a search query or preferred tech stack, the system will process and display relevant repositories from the dataset or GitHub API. The recommendation engine will sort the repositories by the relevance score. The frontend will display the recommended repositories along with key information such as project name, description, language used, stars, forks, issues and links to the GitHub. This allows users to view the recommendations and find appropriate opportunities that they can contribute to open source

4 Implementation Details & Tech Stack

Component	Technology Used	Purpose
Frontend	React.j s	Building the interactive user interface
Styling	Tailwind CSS	Responsive and modern UI design
Programming Language	Python	Backend and recommendation logic
Backend Framework	FastAPI	Developing REST APIs
Repository Data	GitHub REST API	Retrieving repository information
Database	PostgreSQL	Storing user preferences and saved projects
Data Processing	Pandas	Processing and analyzing repository metadata
Recommendation	Scikit-learn	Implementing content-based recommendation techniques
Authentication	JWT / OAuth	Secure user authentication
Deployment	Vercel / Render	Hosting the application

4.1 Sample Recommendation Logic

Repository Name	Language	Stars	Difficulty
FreeCode Camp	Java script	430000	Beginner
scikit-learn	Python	65000	Intermediate
React	JavaScript	240000	Intermediate
TensorFlow	Python	190000	Advanced
VS Code	Typescript	175000	Advanced

4.2 Sample Recommendation Logic

The recommendation engine can calculate a matching score using the following weighted formula:
Recommendation Score:
Score = (Language Match × 0.35) + (Topic Match × 0.30) + (Skill Match × 0.20) +
(Activity Match × 0.15)
Each matching component is normalized between 0 and 1.
The system ranks repositories according to their calculated scores and displays the most relevant projects first.

5 Results and Testing

The proposed Open Source Project Finder is designed to simplify the repository discovery process by providing personalized search results based on user preferences.
The system can be evaluated using different test scenarios, including:

- Searching for repositories using programming languages.
- Filtering projects based on technologies and topics.
- Testing recommendations for different skill levels.
- Verifying repository information retrieved through GitHub.
- Checking the accuracy and relevance of recommendation results.
- Testing authentication and saved-project functionality.

Evaluation Metrics:

1. Recommendation relevance.
2. Search response time.
3. API response success rate.
4. Repository data completeness.
5. User satisfaction and usability.

Actual numerical results should be added after executing the application and conducting the tests.

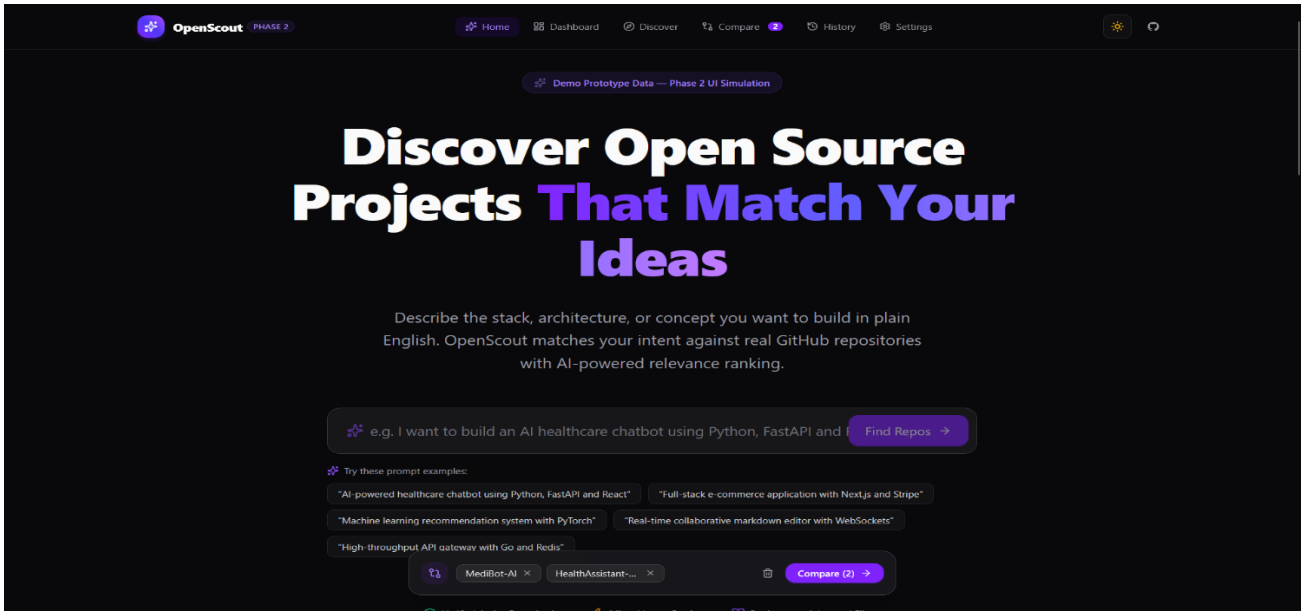


Figure 4: Home Page of Open Source Project Finder

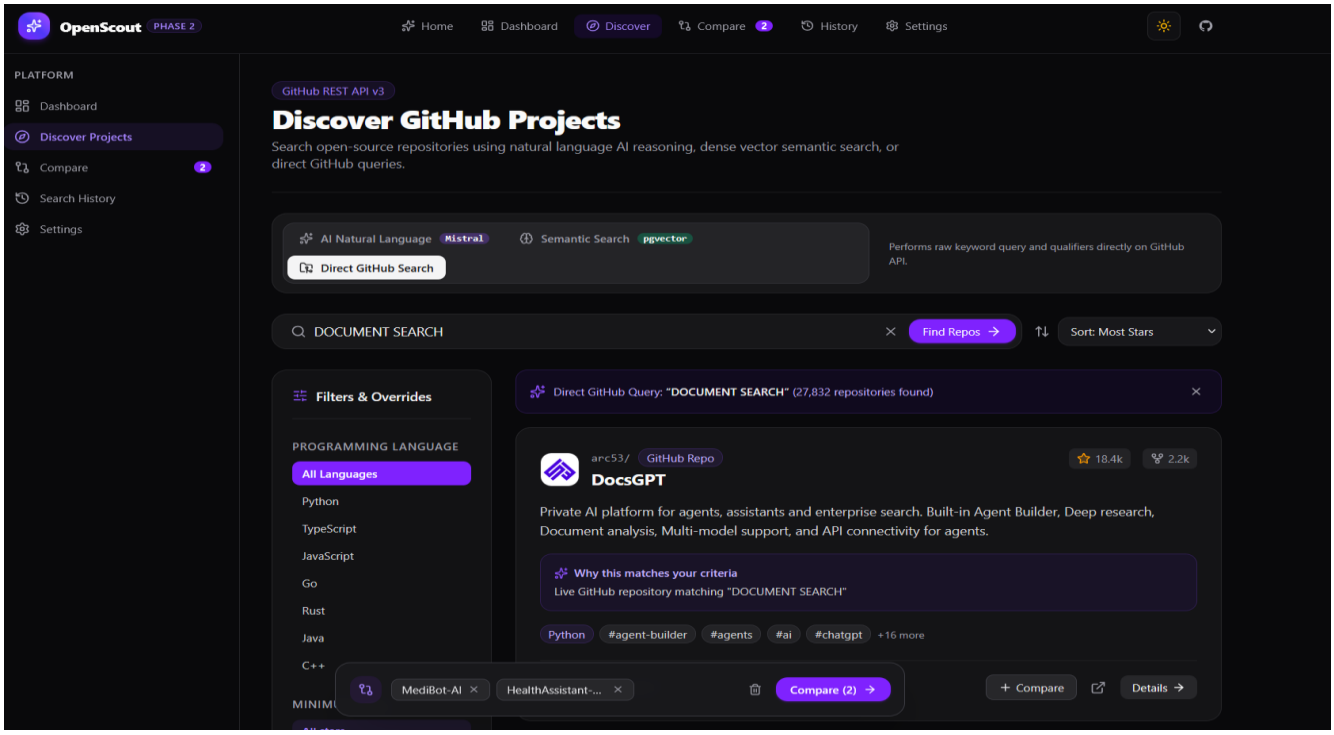


Figure 5: Recommended Open-Source Projects

6 Conclusion & Future Scope

Conclusion: The Open Source Project Finder is built to make it easier to find open source projects by using both repository data from GitHub and custom search and recommendation methods. The platform makes it easier for developers to find relevant projects based on their programming skills and experience. It also reduces the need to search repositories manually.

Future Scope:

- AI-based recommendations for projects tailored to you.
- Integrated with GitHub OAuth to create a personalised profile.
- Identify easy issues to contribute to on GitHub using labelling.
- AI based assessment of how difficult a repo is and what is required to contribute.
- Analysis of skill gaps with personal recommendations for learning.
- Keep track of your contributions and favourite repos.
- Receive e-mails when new issues appear or a repo is updated.
- Integrate with more developer platforms.